

# NETWORKED CONTROL

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## EXERCIZES

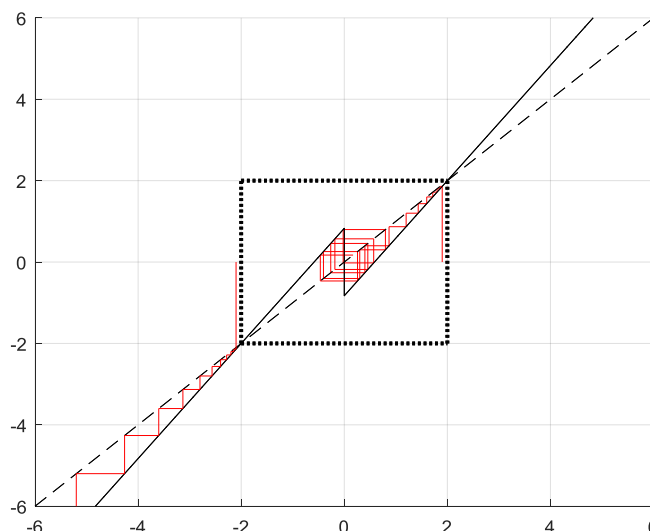
1) Consider the following continuous-time system  $\mathcal{S}$ .

$$\dot{x}(t) = (\log_e 4)x(t) + 2(\log_e 4)u(t)$$

- A. Compute the eigenvalue and the mode of the system. Is the system open-loop asymptotically stable?
- B. We aim to control the system using a digital quantized controller. Answer the following questions.
  - a. According to the theory, what is the minimum rate  $N_B/h$  (where  $N_B$  is the number of bits encoding the control law and  $h$  is the sampling time) required to make the system *boundable*?
  - b. In view of the answer given to the previous question, what is the maximum value of  $h$  that can be used to make the system *boundable* in case  $N_B = 1$ , i.e., with a binary control law?
  - c. Set  $h = 0.25$  s. Using a binary (i.e., where  $N_B = 1$ ) control law of the type  $u_k = -\text{sign}(x_k)$  analyze the properties of the state trajectories using the graphical method (possibly starting by different selected initial conditions).
  - d. In case  $N_B = 1$  and  $h = 0.25$  s, what is the corresponding invariant set?

## SOLUTION

- A. The eigenvalue of the system is  $\log_e 4 > 0$ , making the system open-loop unstable. The corresponding mode is  $e^{(\log_e 4)t} = 4^t$ .
- B. Regarding the digital quantized control system:
  - a. The minimum value of  $N_B/h$  required to make the system *boundable* is  $\log_e 4 \cdot \log_2 e = 2$ .
  - b. We compute that, if  $N_B = 1$ ,  $h_{\max} = 0.5$  s.
  - c. In case  $N_B = 1$  and  $h = 0.25$  the system is boundable, as witnessed by the following figure.



In particular, if  $x_0 \in [-2, 2]$ , then  $x_k \in [-2, 2]$  for all  $k \geq 0$ . On the other hand, if  $x_0 \notin [-2, 2]$ , then  $|x_k| \rightarrow +\infty$  as  $k \rightarrow +\infty$ .

- d. In case  $N_B = 1$  and  $h = 0.25$  s, as witnessed by the figure above, the invariant set is  $\mathbb{I} = [\bar{x}_0, \bar{x}_2]$ , where  $\bar{x}_0 = -\frac{b}{a}u_{\max}$  and  $\bar{x}_2 = -\frac{b}{a}u_{\max} + \frac{b}{a}(u_{\max} - u_{\min})\frac{2}{N}$ . Since  $u_{\min} = -1$ ,  $u_{\max} = 1$ ,  $N = 2$ ,  $a = \log_e 4$ , and  $b = 2 \cdot (\log_e 4)$ , then  $\mathbb{I} = [-2, 2]$  as expected.

2) Assume that the linear system

$$\mathcal{S}: \begin{cases} \dot{x}(t) &= Ax(t) + Bu(t) \\ y(t) &= Cx(t) \end{cases}$$

is controlled through the networked control system (NCS) sketched in the following figure.

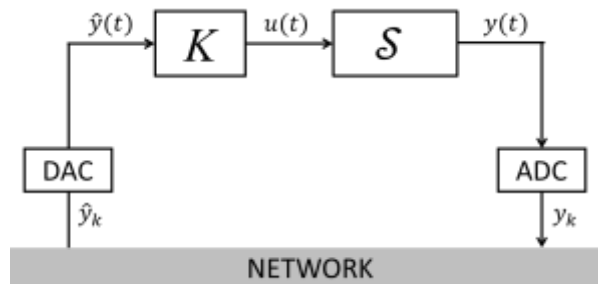


Figure 1: Networked control system

Assume that the sampling time  $h_k$  and the delay  $\tau_k$  affecting the control network are time varying and that  $h_{\min} \leq h_k \leq h_{\max}$ ,  $\tau_{\min} \leq \tau_k \leq \tau_{\max}$ . **State the main theorem providing the condition guaranteeing the exponential stability of the NCS.**

Also, **comment of the numerical solvability of the condition** provided in the theorem and on the **conservativeness of the result**.

**SOLUTION:** See the lecture notes.

3) What are the main differences between MAC protocols based on **scheduling** and the ones based on **random multiple access**?

Also, **list some main protocols** that belong to these two classes (at least one for each class) and briefly **describe their main features**.

**SOLUTION:** See the lecture notes.